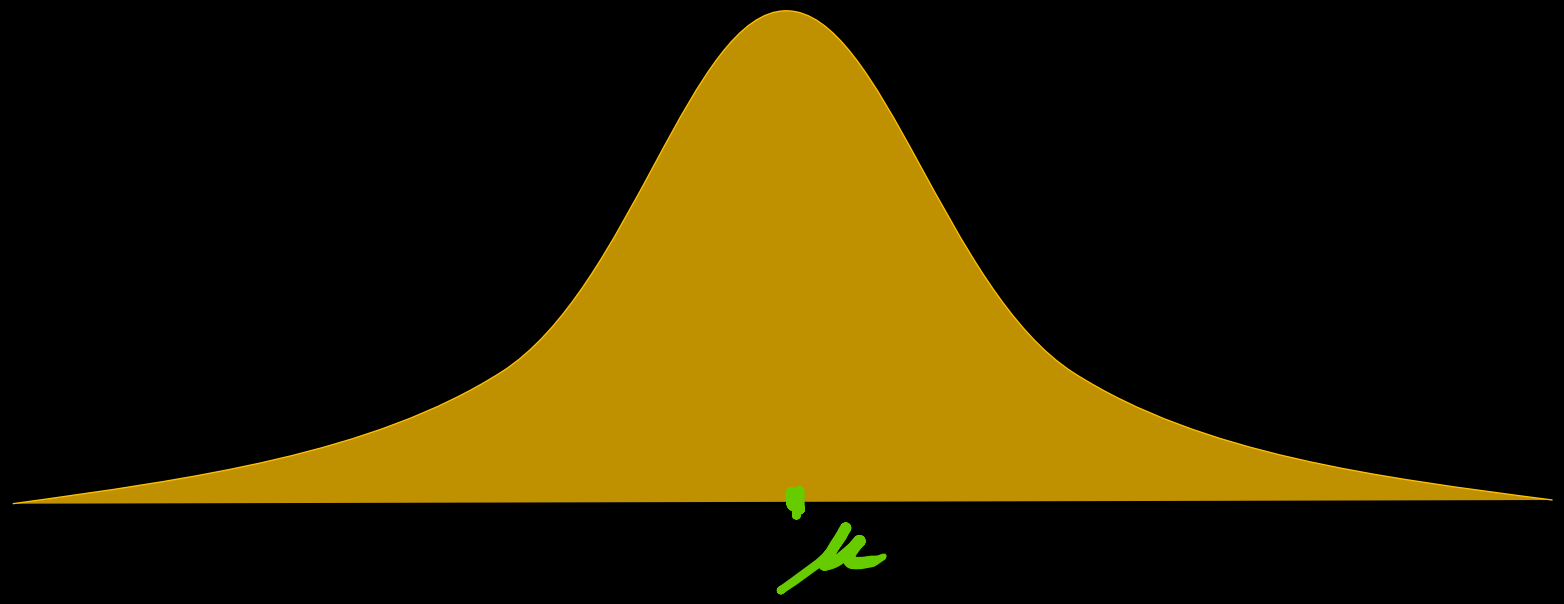
Normal Distribution:

A **normal distribution** is a **continuous probability distribution** in which data is **symmetrically distributed around the mean**, forming a **bell-shaped curve**.

Most values occur near the average (mean), and fewer values appear as you move away from the center.



Suppose you have following data (age):

51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

Suppose we create 2 bins of equal size. We would get following:

Here **bin_width = (max_value - min_value)/num_of_bins = (74-51)/2 = 11.5.**

- The bin starts from minimum value, which is 51, and then increase by 11.5 units.
- The bins go from 51 -> 62.5 -> 74

So the
bin1 : 51 - 62.5
bin2 : 62.5 - 74
We would get following counts for these 2 bins: 36 and 64

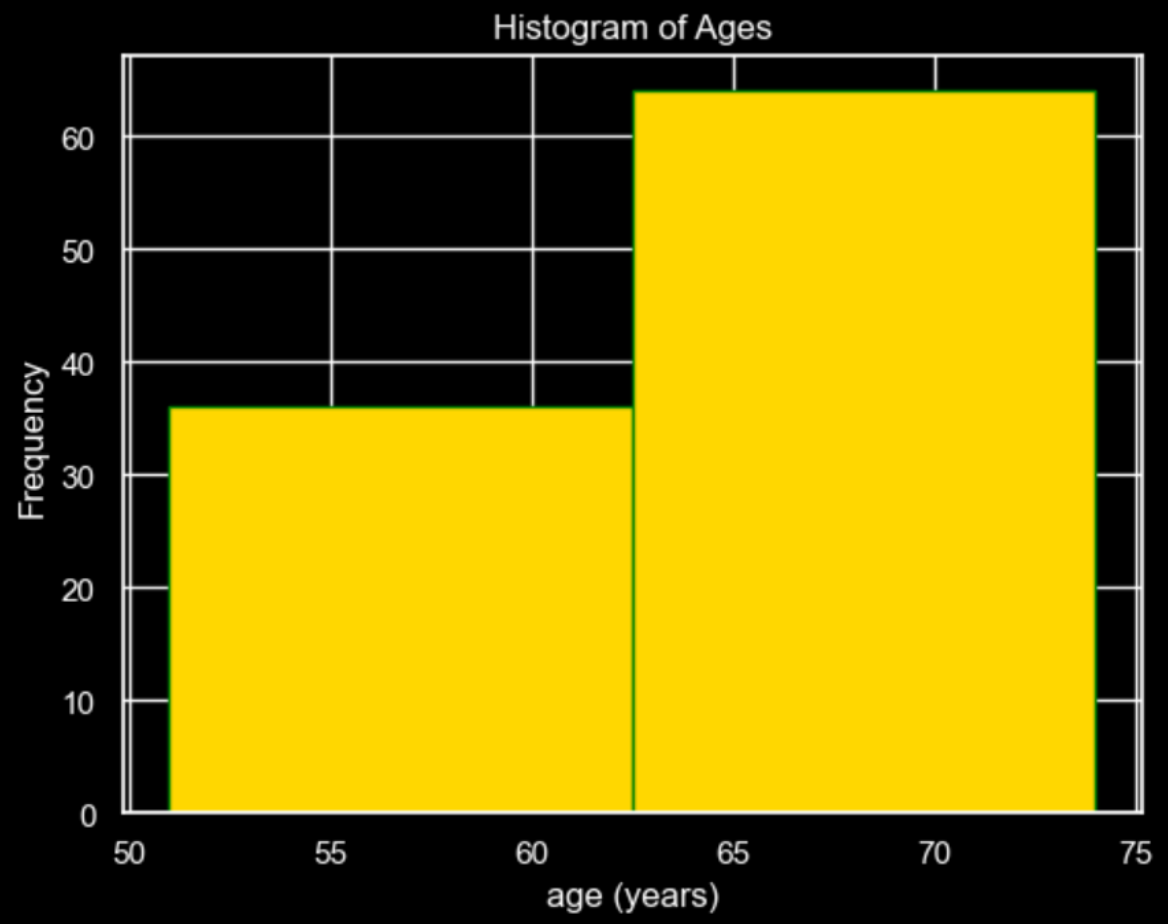
51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

If we create histogram of these 2 bins:

bin1 : 51 - 62.5

bin2 : 62.5 - 74

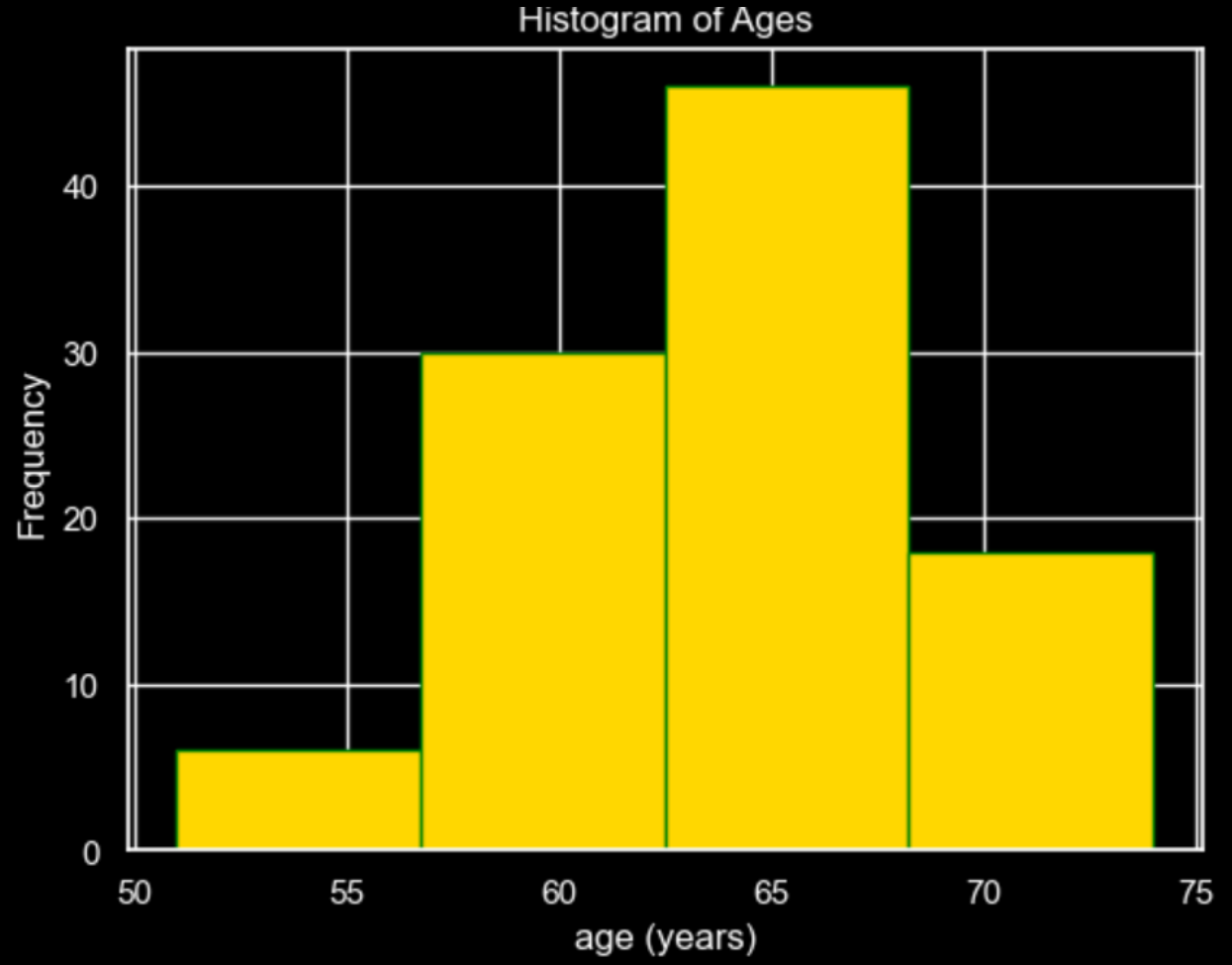
with counts 36 and 64 we would get ->



51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

Let's increase the bin count to 4:

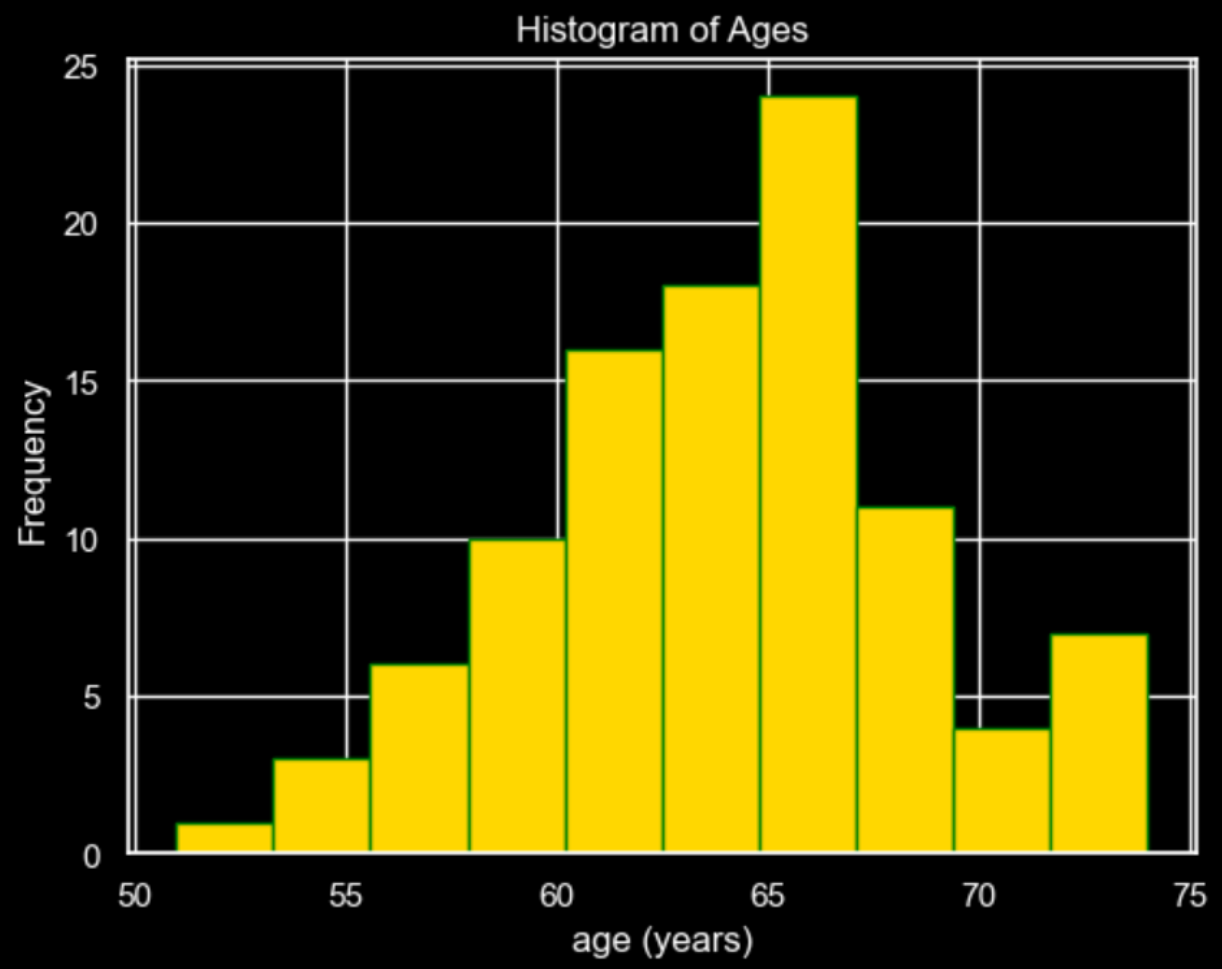
So, the bin width = (74-51)/4 = 5.75
bin1 : 51 - 56.75
bin2 : 56.75 - 62.5
bin3 : 62.5 - 68.25
bin4 : 68.25 - 74
We would get following counts for the 4 bins: 6, 30, 46, and 18



51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

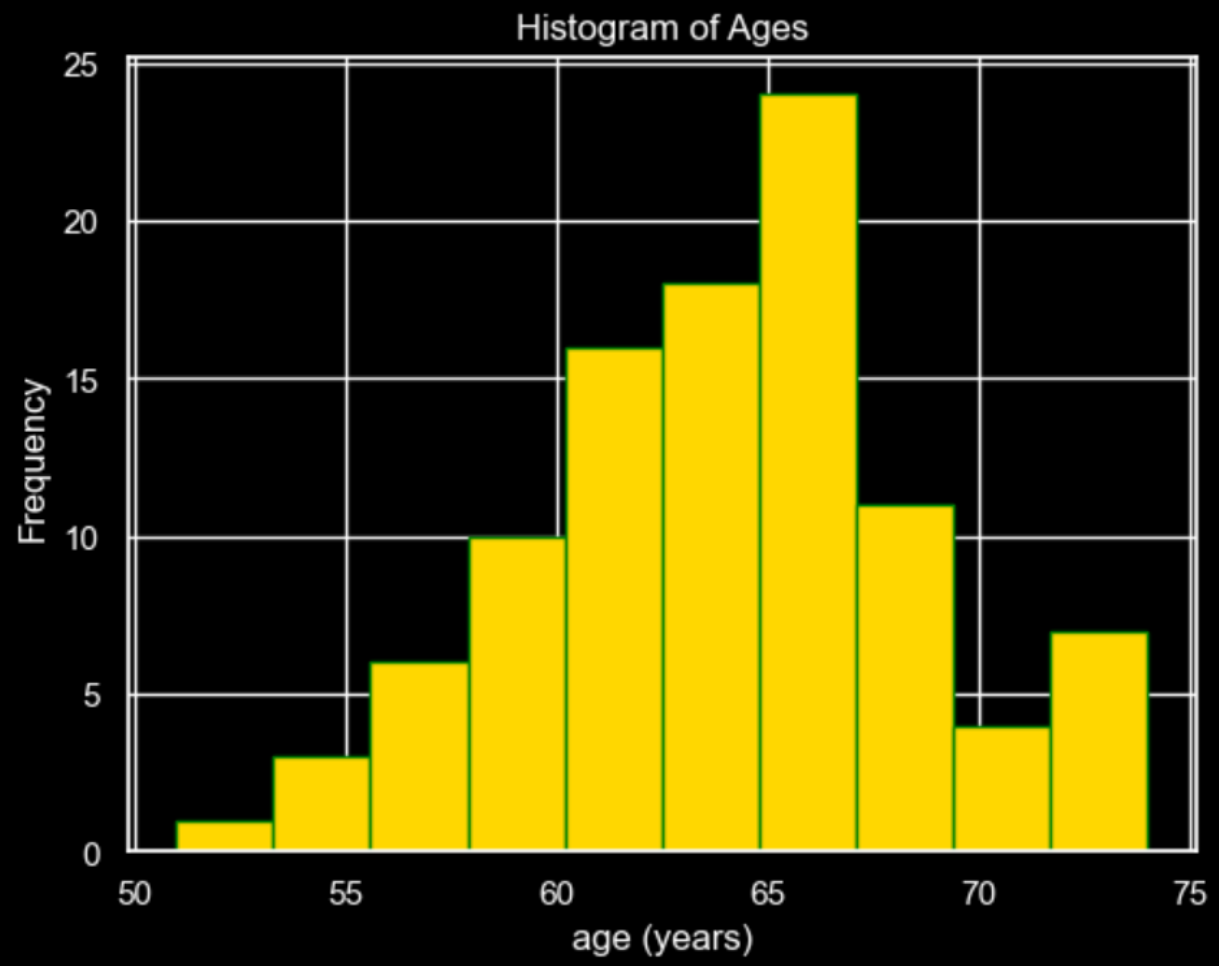
Let's increase the bin count to 10:

So, the bin width = (74-51)/10 = 2.3
bin1 : 51 - 53.3
bin2 : 53.3 - 55.6
....
bin10: 71.7 - 74
We would get following counts for the 10 bins: 1, 3, 6, 10, 16, 18, 24, 11, 4, 7



51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

As bins increases, the histogram becomes bell shaped, that is takes a shape of normal distribution.



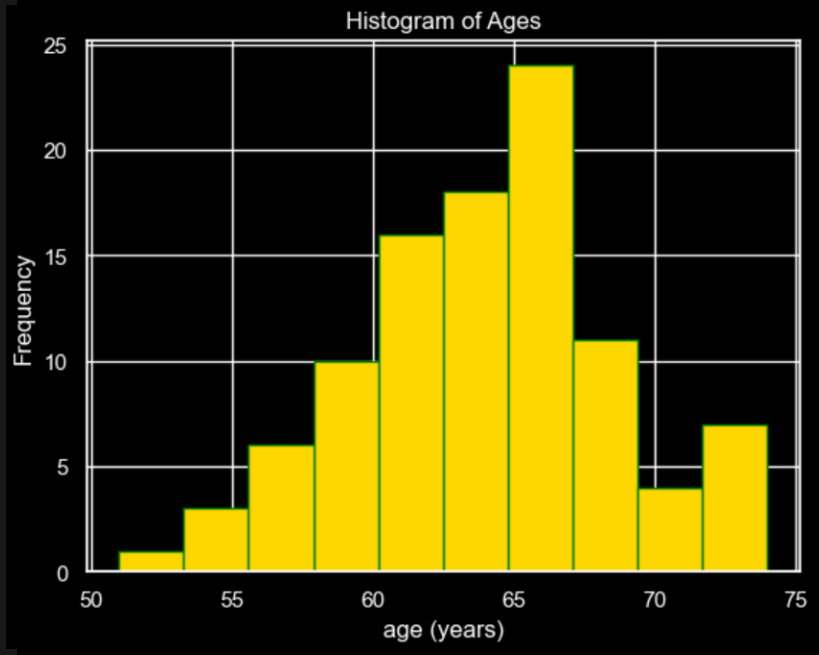
51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59, 59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62, 62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64, 64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74

Code

```
# 1) Generate data
import numpy as np

# Following could be age of employees in a company
data = np.array([51, 55, 55, 55, 56, 56, 57, 57, 57, 57, 58, 58, 59, 59, 59,
    59, 59, 60, 60, 60, 61, 61, 61, 61, 61, 61, 62, 62, 62, 62, 62, 62, 62,
    62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 63, 64, 64, 64,
    64, 64, 64, 65, 65, 65, 65, 65, 65, 65, 65, 66, 66, 66, 66, 66, 66, 66,
    66, 66, 66, 66, 66, 66, 67, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 69,
    69, 69, 70, 70, 70, 71, 72, 72, 72, 72, 72, 72, 74]
)

data = np.sort(data.astype(int))
```

Code

```python
# 3)  Lets increase the bin count to 10
import matplotlib.pyplot as plt

bin_count = 10
counts, bins, patches = plt.hist( data, bins=bin_count,
    edgecolor='green',          # green edges for contrast
    color='gold'                # brighter color for visibility
)

plt.title('Histogram of Ages', color='white')
plt.xlabel('age (years)', color='white')
plt.ylabel('Frequency', color='white')
plt.tick_params(colors='white')
plt.show()
```



Normal Distribution Functions